

Pearl Path Web Application

User Manual & Operating Guide

CO2060 – Software Systems Design Project

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1. About Pearl Path

Pearl Path is a web-based digital travel platform designed to make travel planning in Sri Lanka easier by bringing several travel-related services into one unified application.

The platform provides centralized access to:

- Accommodation services
- Vehicle rental services
- Tour guide services
- Advanced service search and multi-criteria filtering
- Booking and real-time availability management
- Interactive route planning
- Dynamic currency conversion
- AI-based travel assistance

Pearl Path is designed for three main types of users:

- Tourists
- Service Providers
- Administrators

This manual explains how each type of user can access and use the functions provided by the system.

2. User Roles

Pearl Path uses role-based access control (RBAC) so that users can access the specific functions relevant to their responsibilities.

User Role	Main Functions
Tourist	Search services, view listings, plan routes, make bookings, use currency conversion, and access AI travel assistance.
Service Provider	Manage service listings, update operational availability, and manage customer booking requests.
Administrator	Manage user accounts, verify service provider submissions, and monitor platform activity and security.

Note: The available screens, navigation options, and active permissions will differ depending on the user's logged-in account role.

3. System Requirements

Pearl Path is a responsive web application that can be accessed from desktop computers, tablets, and mobile devices.

3.1 Supported Browsers

A recent version of one of the following web browsers is recommended for optimal performance and security:

- Google Chrome
- Mozilla Firefox
- Microsoft Edge
- Apple Safari

3.2 Internet Connection

A stable internet connection is recommended. An active internet connection is required for features that depend on external live services, including:

- Interactive map rendering and route calculations
- Email-based OTP verification and transactional emails
- Dynamic exchange-rate currency conversion
- AI travel assistant query processing

3.3 Browser Settings

JavaScript must be enabled in your browser settings to allow dynamic UI components and backend communication.

4. Getting Started

4.1 Accessing Pearl Path

1. Open a supported web browser.
2. Navigate to the Pearl Path web application URL.
3. The Pearl Path home page will be displayed.
4. Use the top navigation bar to browse available sections.
5. Select **Register** to create a new account or **Login** to access an existing account.

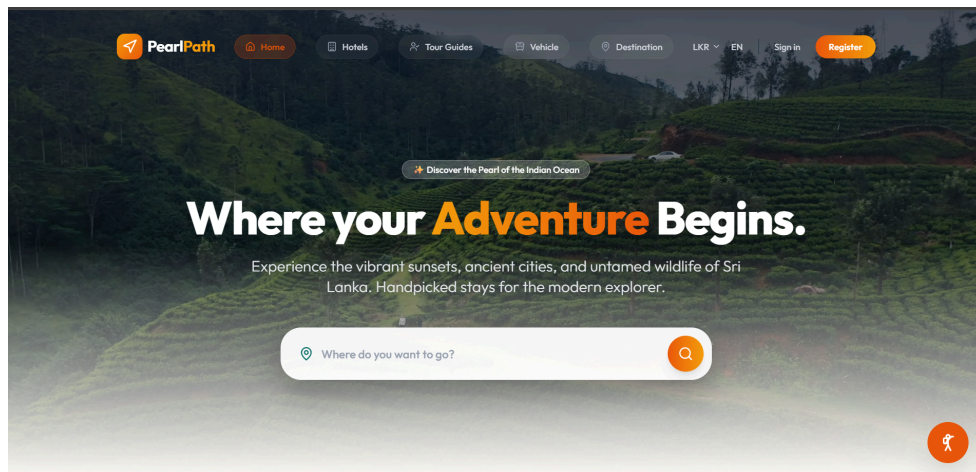


Figure 1: Pearl Path Home Page

4.2 Creating an Account

Users must create an account before accessing features that require authentication.

Steps

1. Click **Register** on the top navigation bar.
2. Select the required account type:
 - **Tourist**
 - **Service Provider**
3. Enter the required registration details.
4. Provide a valid email address.
5. Create a secure password.
6. Click **Sign Up**.
7. The system will dispatch a verification One-Time Password (OTP) to the registered email address.

Important: Always use an accessible email address, as it is strictly required for account verification and password recovery.

Figure 2: Registration Page

4.3 OTP Verification

After registration, the account must be verified using the OTP sent to the registered email address.

Steps

1. Open the verification email received in your inbox.
2. Copy the 6-digit OTP code.
3. Enter the OTP on the Pearl Path verification screen.
4. Click **Verify Account**.
5. After successful verification, your account will be activated for immediate login.

If the OTP is not received

1. Check your **Spam** or **Junk** folder.
2. Confirm that the email address entered during registration was correct.
3. Wait briefly for network delivery.
4. Select **Resend OTP** if the option becomes active.

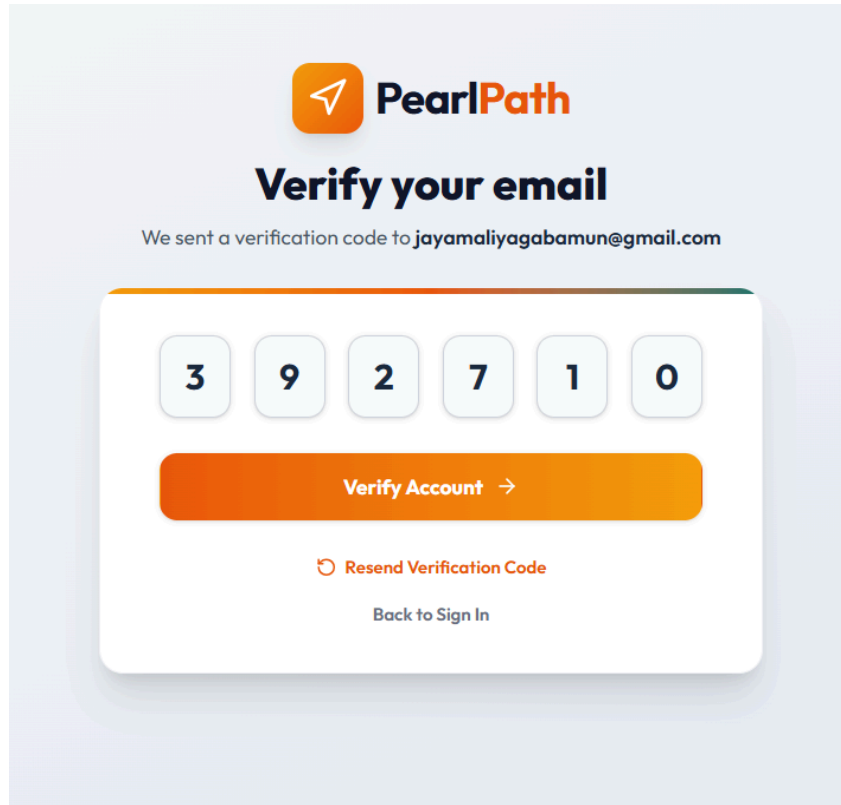


Figure 3: OTP Verification

4.4 Logging In

Steps

1. Click **Login** on the main navigation bar.
2. Enter your registered email address.
3. Enter your account password.
4. Click **Log In**.
5. Upon successful authentication, the system redirects you to your relevant user dashboard.

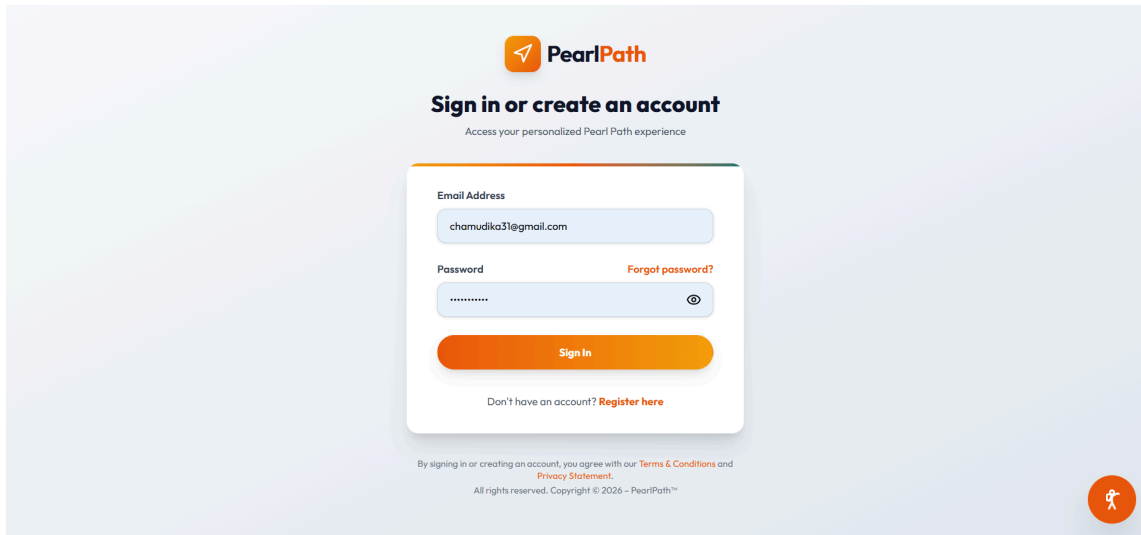


Figure 4: Login Page

4.5 Recovering a Forgotten Password

If you forget your password, use the self-service password recovery feature.

Steps

1. Open the **Login** page.
2. Click **Forgot Password?**
3. Enter your registered email address.
4. Follow the verification workflow sent to your inbox.
5. Enter the reset OTP code when prompted.
6. Create and confirm your new password.
7. Return to the login page and sign in using your new credentials.

5. Tourist Guide

5.1 Tourist Dashboard

After signing in with a Tourist account, users gain full access to travel discovery and planning tools.

Key functions available on the Tourist Dashboard:

- Explore available travel services
- Multi-criteria search and filter
- View complete service details and provider profiles
- Plan routes between destinations
- Submit service bookings
- Track and manage submitted reservations
- Convert pricing into local/foreign currencies
- Consult the AI travel assistant

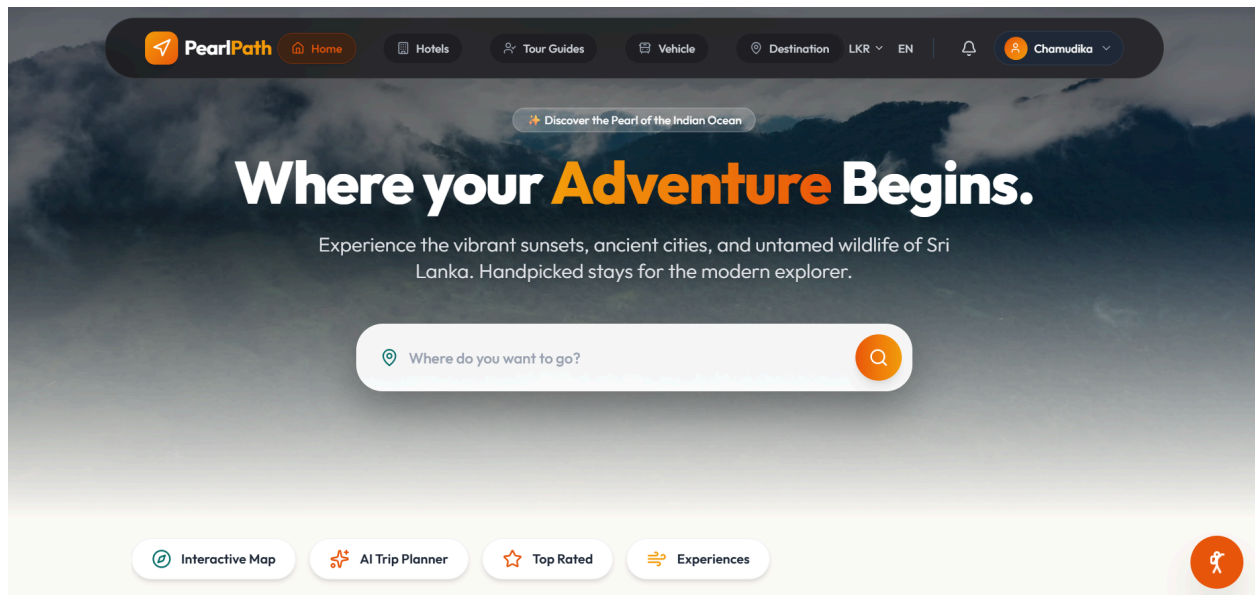


Figure 5: Tourist Interface

5.2 Exploring Travel Services and Experiences

Tourists can explore verified accommodations, tour guides, vehicle rentals, destinations, and curated experiences across Sri Lanka directly from the Pearl Path platform.

Steps

1. Navigate to the top navigation bar or the home screen exploration shortcuts.
2. Select the desired service or exploration category:
 - **Hotels:** Browse verified stays, resorts, villas, and chalets with amenity lists and room options.
 - **Tour Guides:** Find licensed local guides, view spoken languages, experience levels, and daily rates.
 - **Vehicles:** Search transport options including cars, vans, buses, and tuk-tuks with seating and transmission details.
 - **Destinations:** Explore major Sri Lankan travel hubs, regional weather reports, and local attraction summaries.
 - **Experiences:** Discover handpicked activities, cultural tours, and outdoor adventure options.
3. Browse through the resulting listing cards displayed on the page.
4. Click on any service or destination card to view comprehensive details, image galleries, and pricing.

Listing profiles typically display:

- Service name & visual gallery
- Detailed description & amenity highlights
- Geographical location
- Real-time pricing
- Customer ratings & verified reviews

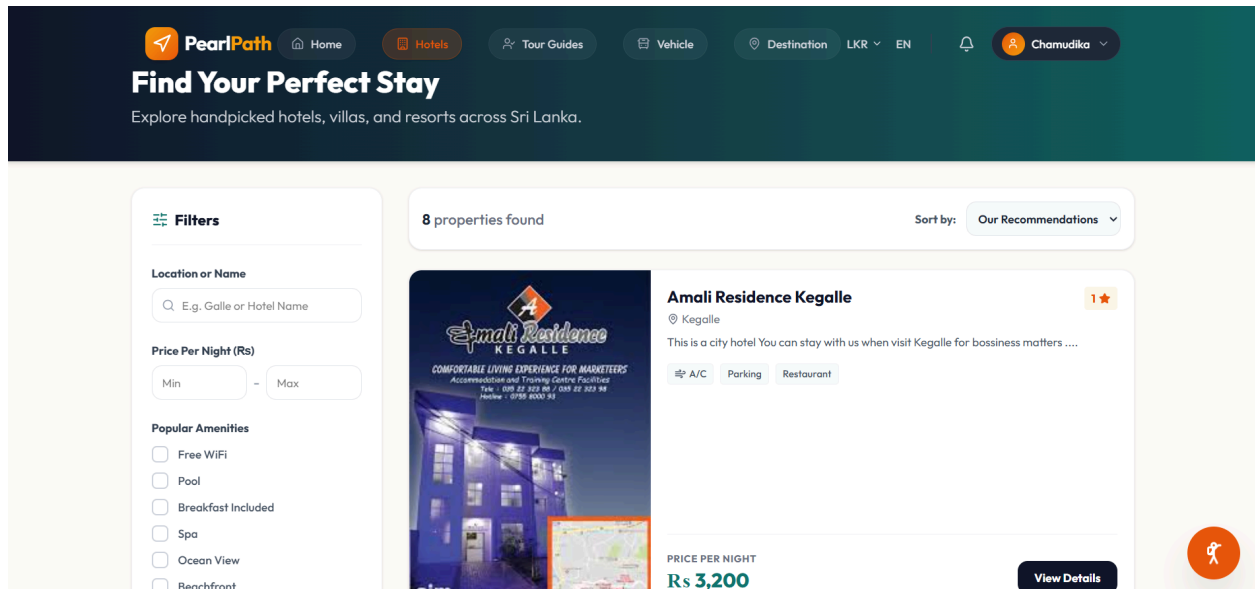


Figure 6: Hotel Services Page

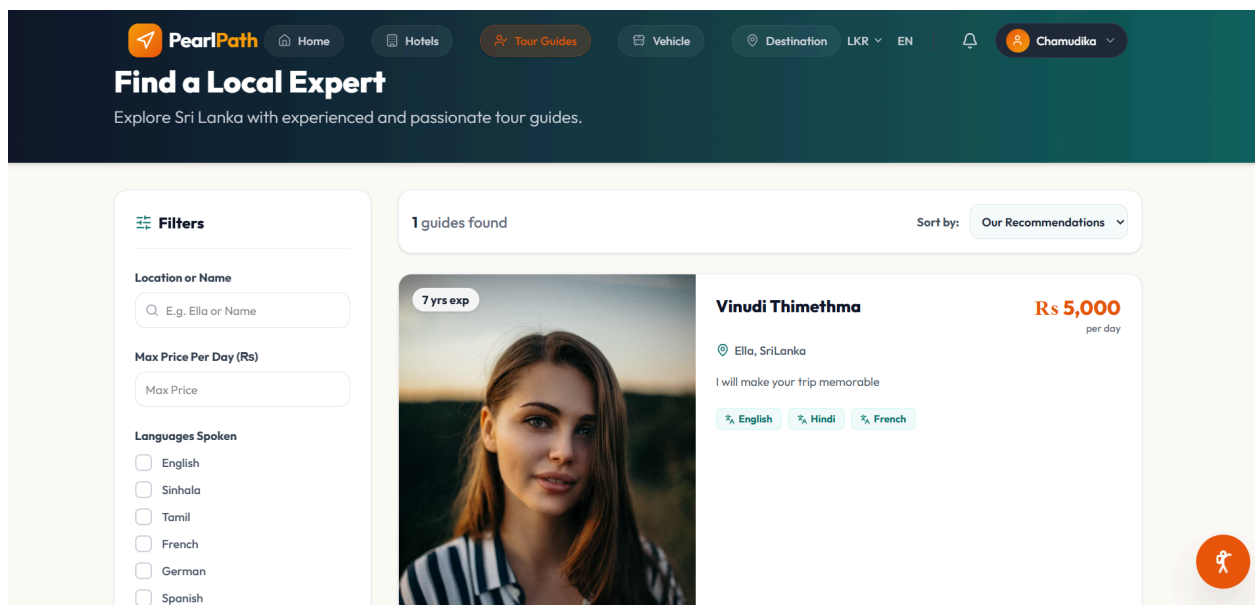


Figure 7: Tour Guide Services Page

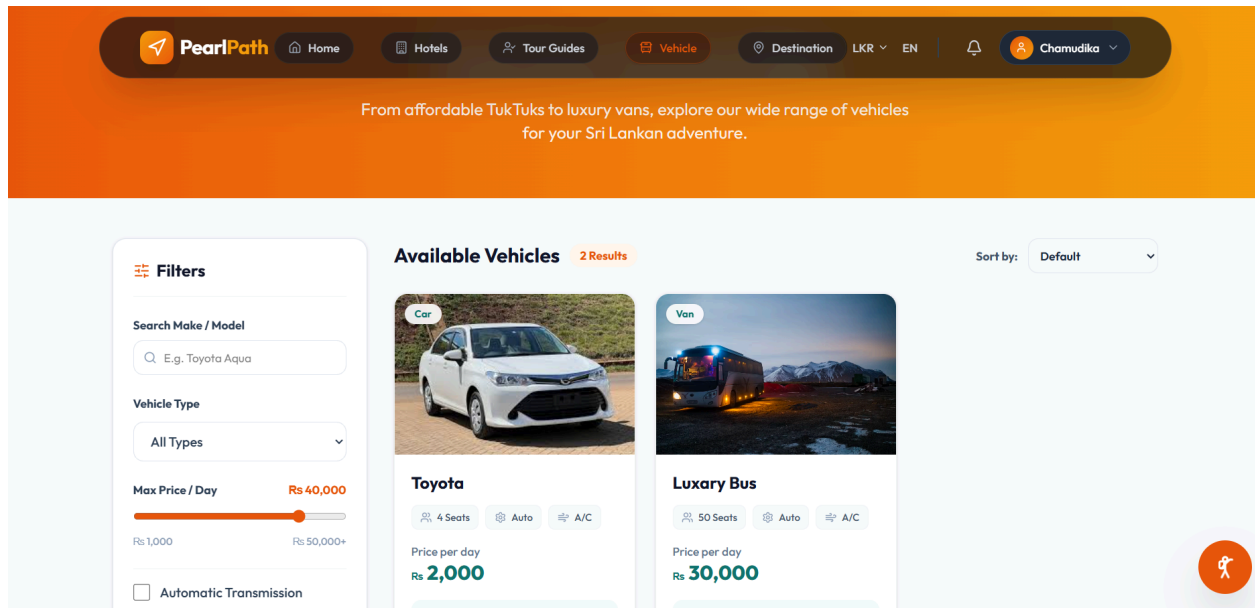


Figure 8: Vehicle Rental Services Page



Figure 9: Destinations Page

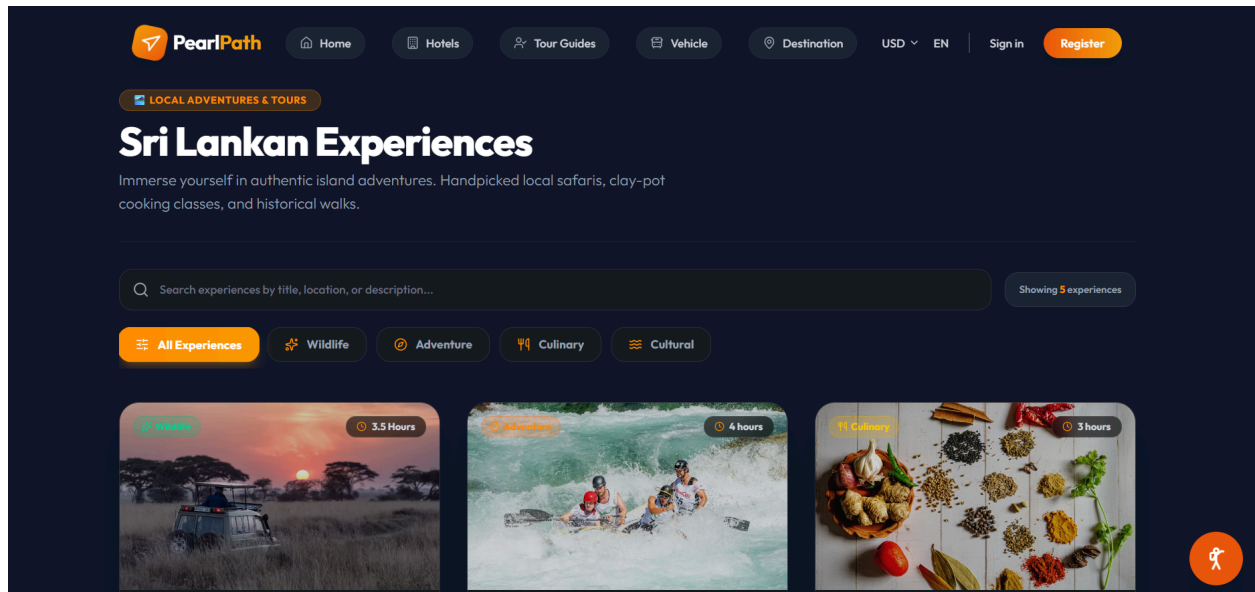


Figure 10: Experiences Page

5.3 Searching and Filtering Services

Use search and multi-criteria filters to refine options based on specific travel parameters.

Steps

1. Navigate to the desired service category.
2. Enter search terms (e.g., location name or service title).
3. Adjust filtering criteria according to your preferences.
4. Review the dynamic search results.
5. Select a listing to view full booking details.

Available filter options:

- Location / Region
- Price range (Min/Max)
- Minimum user rating
- Category-specific tags (e.g., vehicle type, hotel amenities)

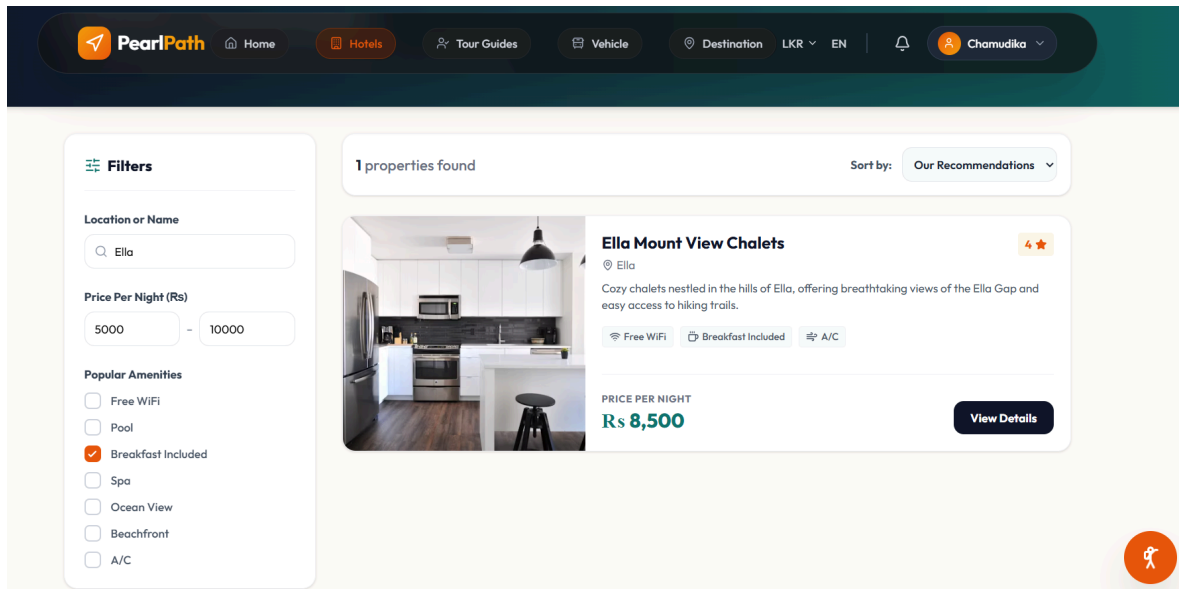


Figure 11: Search and Filter Options

5.4 Viewing Service Details

Before committing to a reservation, tourists can review comprehensive information about any service.

Steps

1. Click on a service listing card from the search grid.
2. Review the image gallery and descriptive overview.
3. Check listed facilities, terms, and specifications.
4. Examine transparent pricing and review breakdowns.
5. Proceed to the booking module if the listing meets your criteria.

5.4.1 Direct Provider Contact

Tourists can communicate directly with verified service providers prior to placing a formal booking to ask questions regarding amenities, availability, or special arrangements.

Steps

1. Open the target service listing page.
2. Locate the **Contact Property / Provider** card on the page.
3. Select your preferred direct contact method:
 - a. **Call Phone**: Click to place a direct call to the provider's registered phone number.
 - b. **Chat on WhatsApp**: Click to open a direct WhatsApp chat window with the provider.
4. Send your inquiry directly to the provider for a quick response.

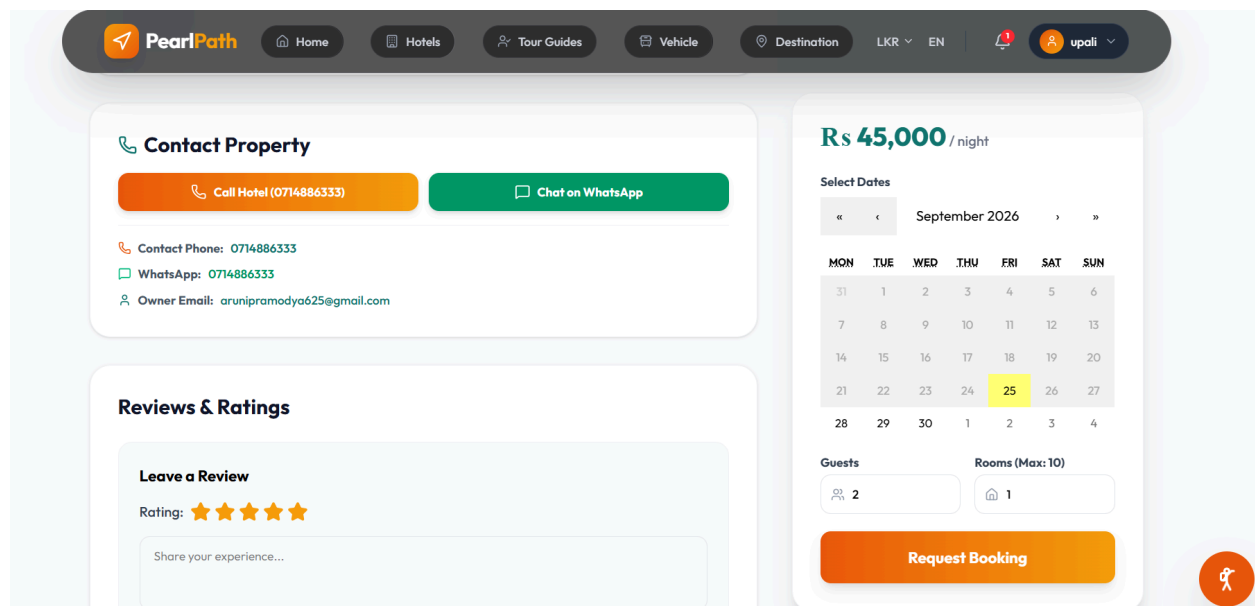


Figure 12: Direct Provider Contact Options

5.5 Currency Conversion

Pearl Path includes live currency conversion to assist international travelers in estimating costs.

Steps

1. Locate the **Currency Switcher** in the navigation header or listing page.
2. Click the dropdown menu.
3. Choose your preferred display currency.
4. All displayed prices across listings will update automatically.

Supported currencies:

- **LKR** – Sri Lankan Rupee
- **USD** – United States Dollar
- **EUR** – Euro
- **GBP** – British Pound

Important: Displayed conversions are estimates based on periodic exchange rate syncs. Final transaction charges may vary depending on your financial institution's conversion rates at billing.

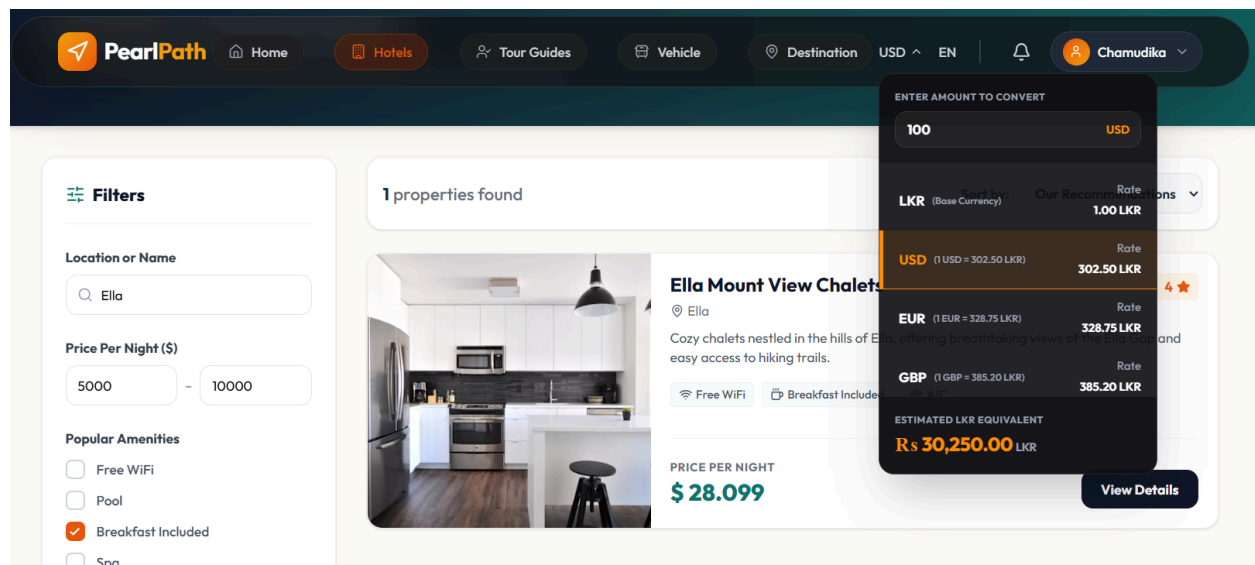


Figure 13: Currency Conversion

5.6 Route Planning

Plan transit routes between Sri Lankan destinations directly within the application using integrated mapping services.

Steps

1. Select **Route Planner** from the navigation menu.
2. Type or select your **Starting Location**.
3. Type or select your **Destination**.
4. Click **Calculate Route**.
5. Review the map output and transit metrics.

The route planner displays:

- Visual route overlays on an interactive map
- Estimated travel distance (km)
- Estimated travel duration
- Waypoint guidance

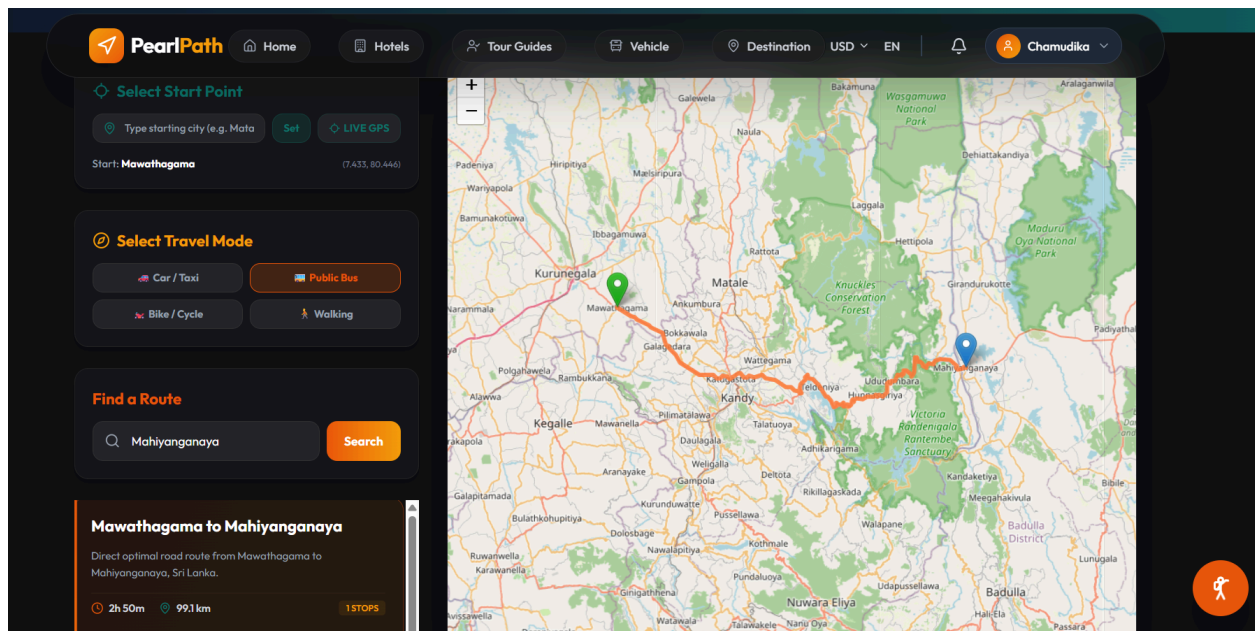


Figure 14: Route Planner

5.7 Booking a Service

Tourists can submit direct reservation requests for available hotel stays, vehicles, or guides.

Steps

1. Open the target service listing.
2. Select your desired start and end dates.
3. Verify total pricing and date selections.
4. Click **Book Now**.
5. The system automatically validates date availability to prevent overlap.
6. Confirm booking summary details.
7. Submit the reservation request.
8. Monitor booking status under your personal profile.

5.7.1 Payment Methods and Payment Slip Upload

To complete and confirm a service booking, tourists must submit payment using supported payment channels and upload proof of payment.

Steps

1. Select desired booking dates and click **Book Now**.
2. On the payment screen, review the accepted payment options (e.g., Bank Transfer / Online Payment Gateway).
3. Transfer the total required amount to the specified bank account or account details provided.
4. Take a clear photo or screenshot of your bank transfer receipt / payment slip.
5. Click **Upload Payment Slip** and select the receipt image file from your device.
6. Click **Submit Booking** to send your payment proof for provider verification.

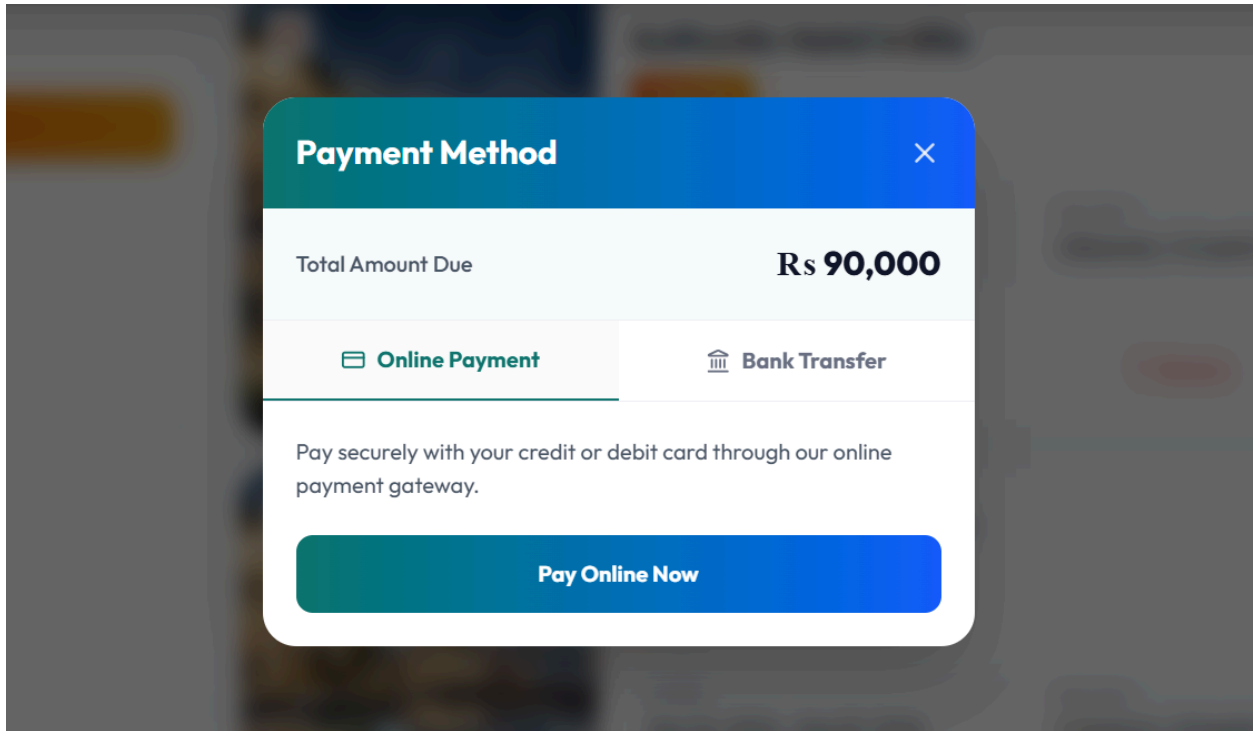


Figure 15: Payment Method Selection

Note: Always ensure the payment slip image is clear and legible. Bookings will remain pending until the service provider or administrator verifies the submitted payment slip.

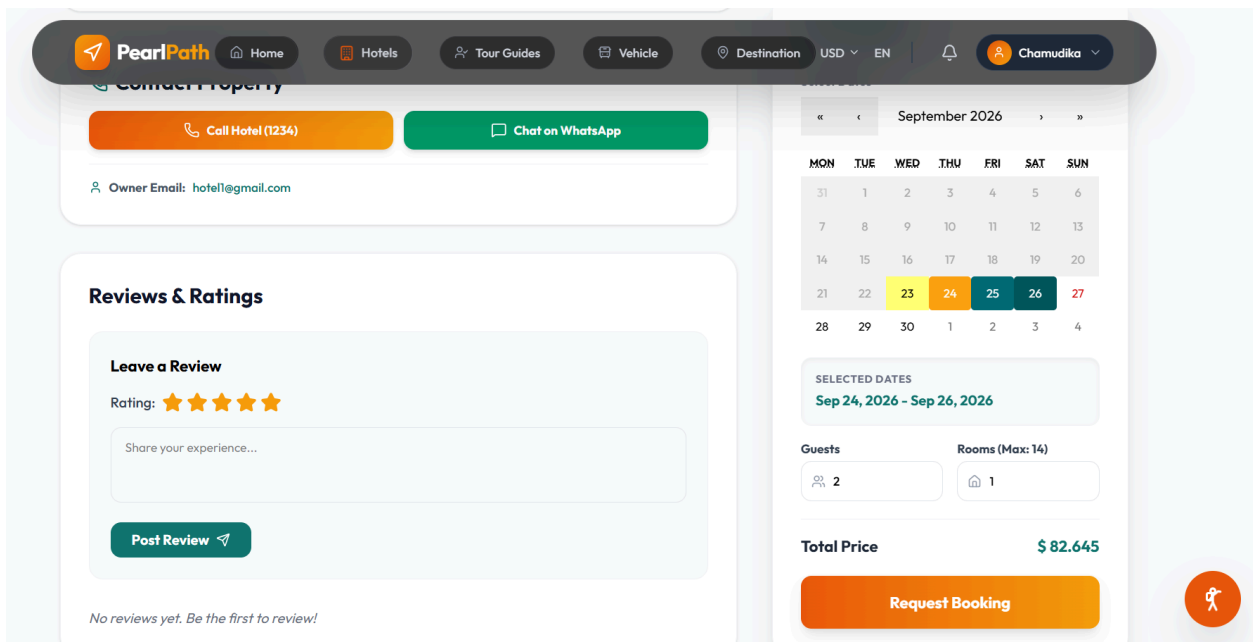


Figure 16: Booking Form

5.8 Managing Bookings

Tourists can monitor current and historical booking requests from their account panel.

Steps

1. Log into your Tourist account.
2. Open **My Account**.
3. Select **My Bookings**.
4. Select any booking entry to view its complete record.
5. Review the status badge and provider details.

Possible booking status states:

- **Pending:** Awaiting review by the service provider.
- **Confirmed:** Approved by the provider.
- **Cancelled:** Reservation declined or cancelled.

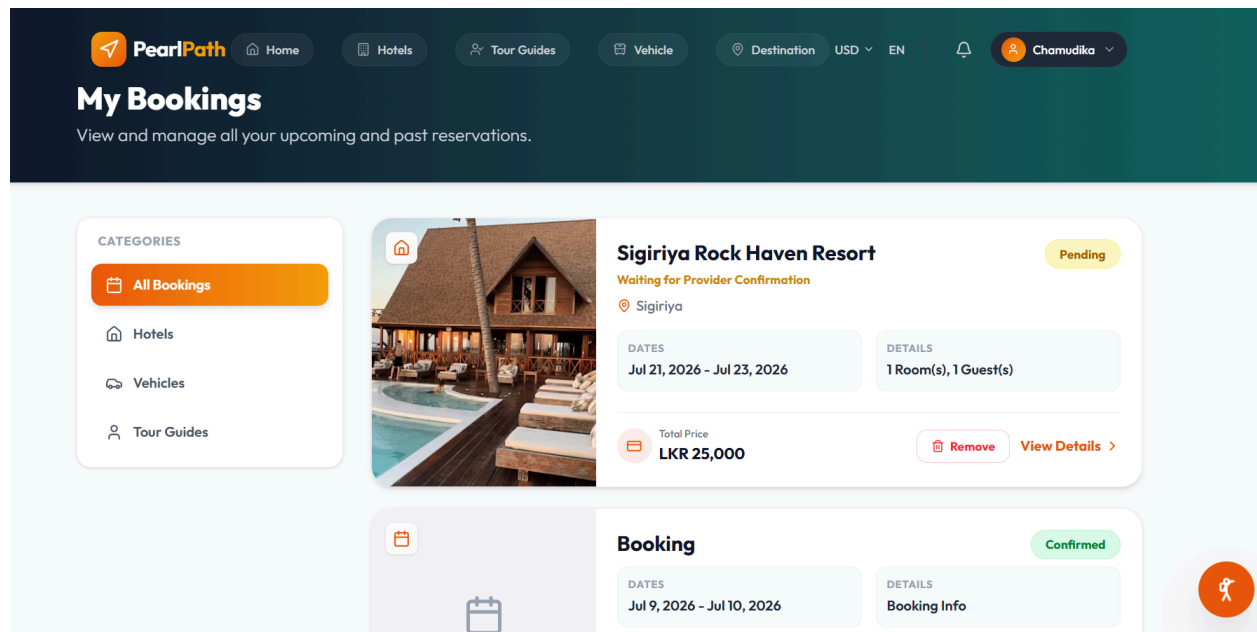


Figure 17: My Bookings

5.8.1 Booking Vouchers and Receipts

Once a booking request is confirmed by the service provider, tourists can access downloadable digital vouchers and billing receipts.

Steps

1. Navigate to **My Account** → **My Bookings**.
2. Locate the target confirmed booking entry.
3. Click **View Details**.
4. Select **Download Voucher / Receipt** to save or print official confirmation documents.

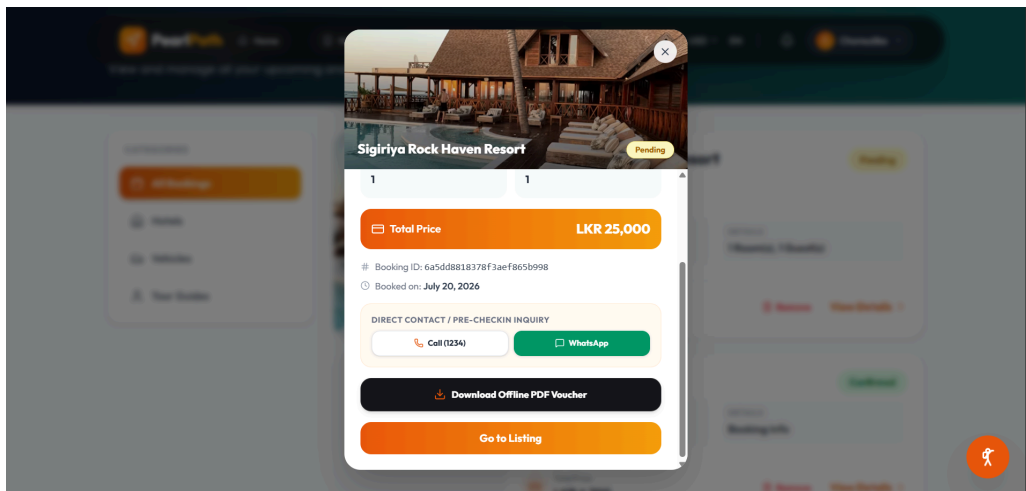


Figure 18: Download Voucher Option

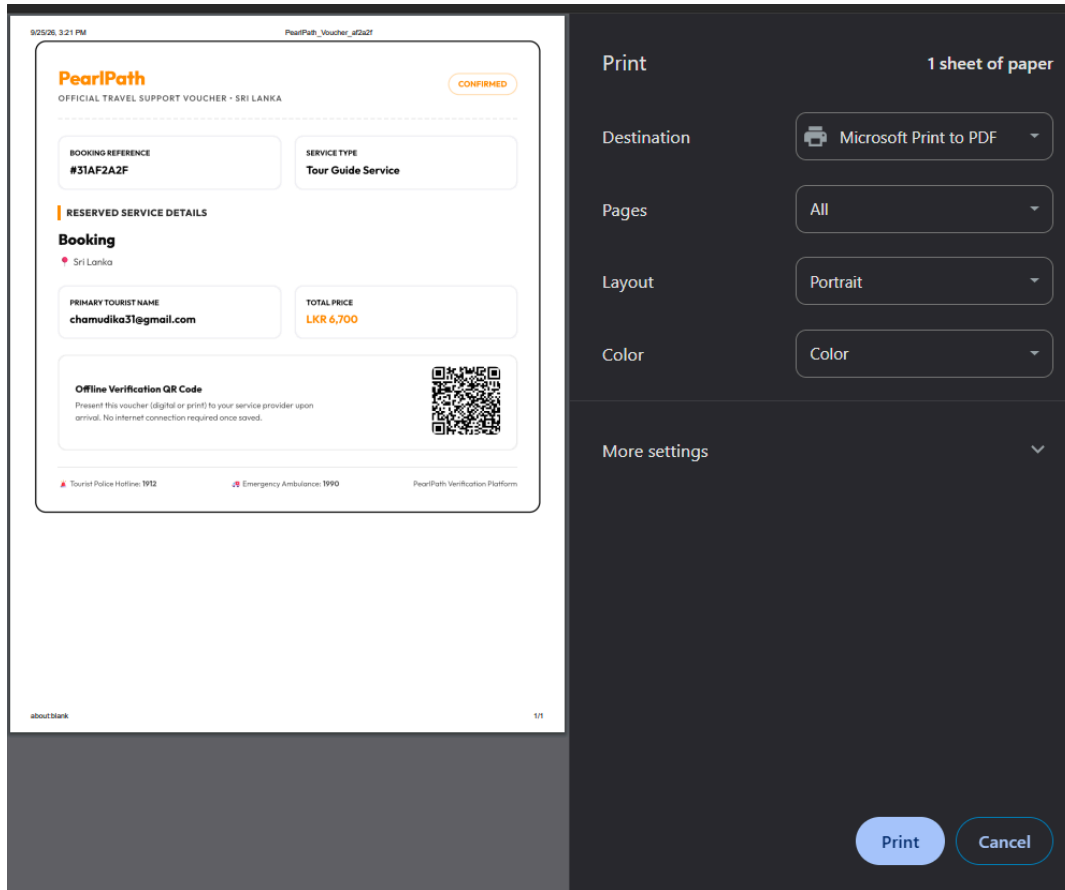


Figure 19: Booking Voucher

5.9 Booking Availability

Pearl Path runs server-side validation against existing database records when dates are selected.

If your chosen dates are already reserved:

1. An availability alert will be displayed.
2. Adjust your start or end dates using the date picker, or
3. Search for alternative verified providers in the same region.

5.10 AI Travel Assistant

An AI chatbot is built into Pearl Path to answer travel queries, recommend destinations, and assist with itinerary ideas.

Steps

1. Click the **AI Chatbot** floating icon in the bottom corner of the screen.
2. Type your travel query into the chat input.
3. Press **Send** or Enter.
4. Review the AI-generated recommendations.

Example prompts:

- *"Recommend budget hotels in Ella."*
- *"What attractions can I visit in Kandy in 2 days?"*
- *"How can I hire a licensed tour guide in Sigiriya?"*
- *"What is the best scenic route from Colombo to Kandy?"*

Important: AI responses provide general recommendations. Users should independently verify critical operational details such as opening hours, seasonal entry fees, and safety notices.

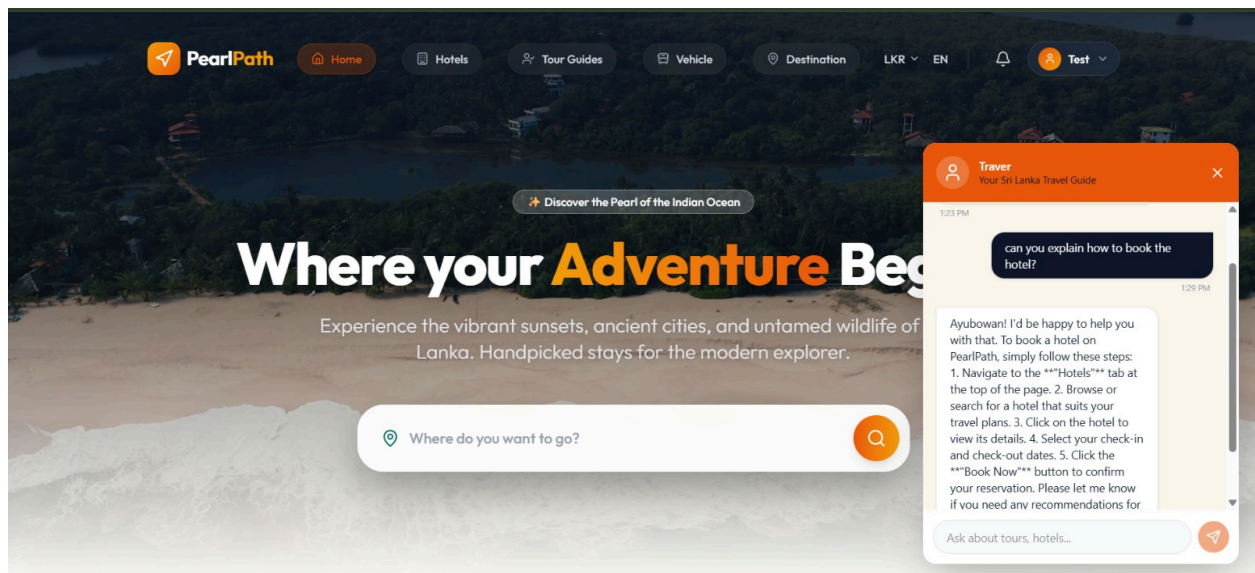


Figure 20: AI Travel Assistant

5.11 AI Trip Planner

In addition to the AI chatbot, Pearl Path includes an automated AI Trip Planner that builds customized multi-day travel itineraries based on traveler preferences.

Steps

1. Click **AI Trip Planner** on the Tourist Dashboard or top navigation bar.
2. Select your travel duration, preferred activity types, and budget range.
3. Click **Generate Itinerary**.
4. Review suggested day-by-day schedules, destination stops, and recommended stays.

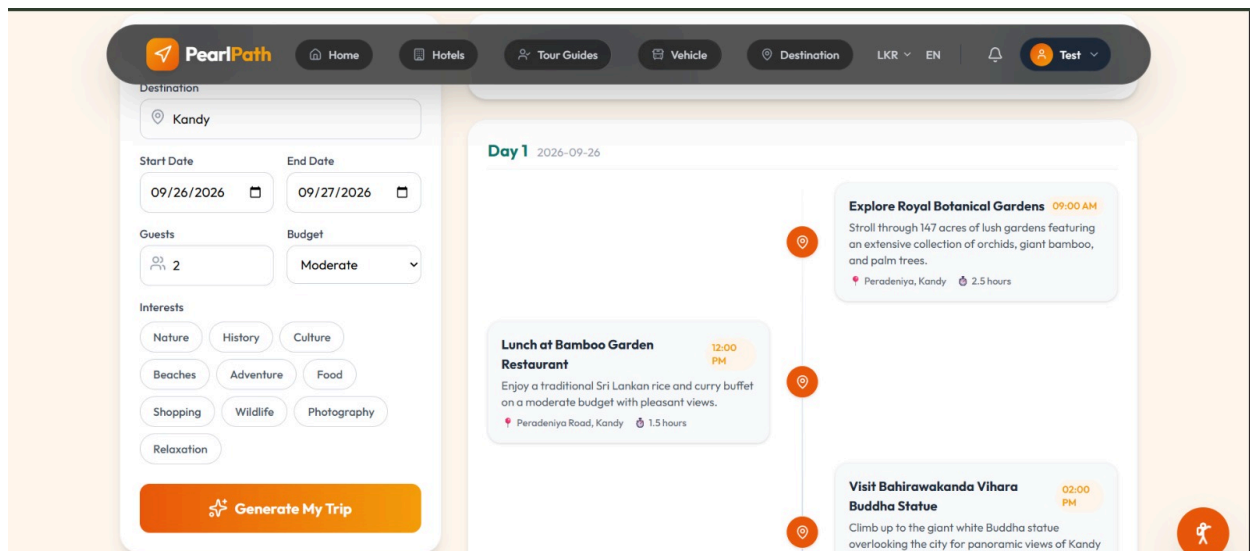


Figure 21: Generated Trip Itinerary

Note: The AI-generated itinerary is intended as a planning aid. Users should verify important information such as service availability, prices, opening times, and other travel details before making arrangements.

5.12 Destination Weather Information

Tourists can inspect real-time weather forecasts and seasonal climate indicators for Sri Lankan travel destinations before making reservations.

Steps

1. Select **Destinations** from the main navigation menu.
2. Choose a region or city (e.g., Ella, Kandy, Sigiriya).
3. View current temperature, weather conditions, and seasonal rainfall guidance.

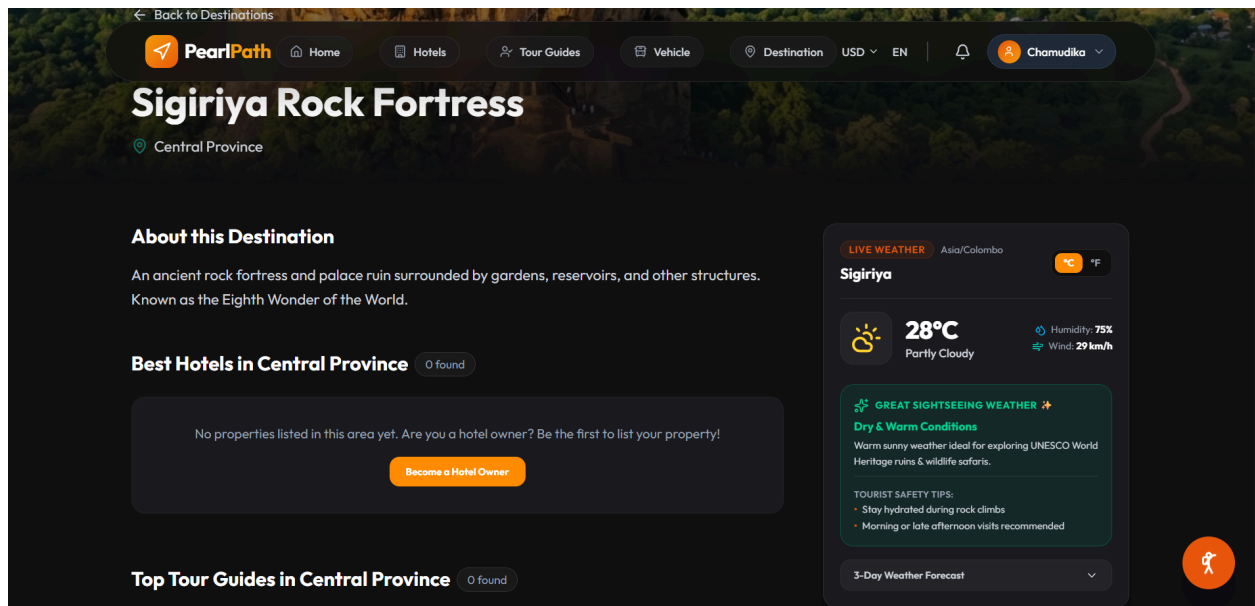


Figure 22: Destination Weather Information

6. Service Provider Guide

6.1 Provider Registration

Hotels, transport operators, and tour guides must register as Service Providers to list services on Pearl Path.

Steps

1. Click **Register** on the main navigation bar.
2. Choose **Service Provider**.
3. Select your provider type (Hotel Owner / Vehicle Owner / Tour Guide).
4. Enter required business and personal details.
5. Provide verification credentials or business license numbers.
6. Submit your registration request.

Note: New service provider accounts require administrative review and verification before listings become publicly visible.

6.2 Provider Verification

To maintain platform trust, administrators review all submitted provider profiles. Providers should upload clear, accurate business documentation during registration to prevent approval delays.

Verification documents may include:

- Official Identity Card / Passport copy
- Business Registration Certificate
- Tourism Authority / Guide License details

6.3 Provider Dashboard

Logged-in service providers access a dedicated dashboard designed for inventory management and customer request handling.

Main Provider Dashboard features:

- Create and manage service listings
- Edit availability schedules and pricing
- Review incoming booking requests
- Update booking statuses (Accept / Decline)

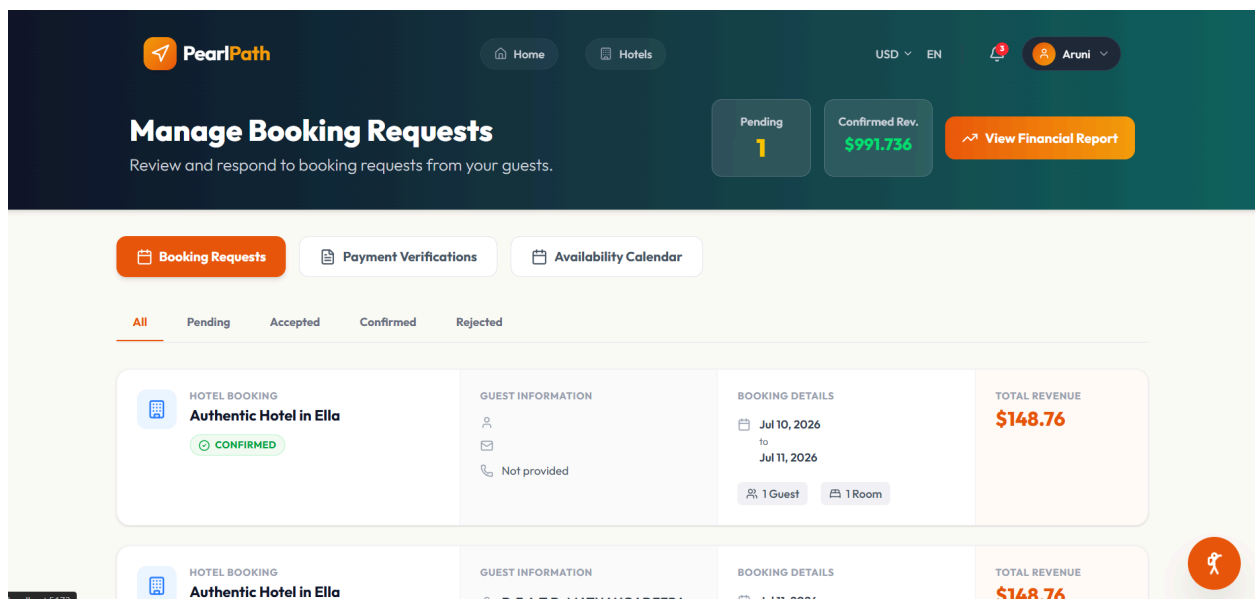


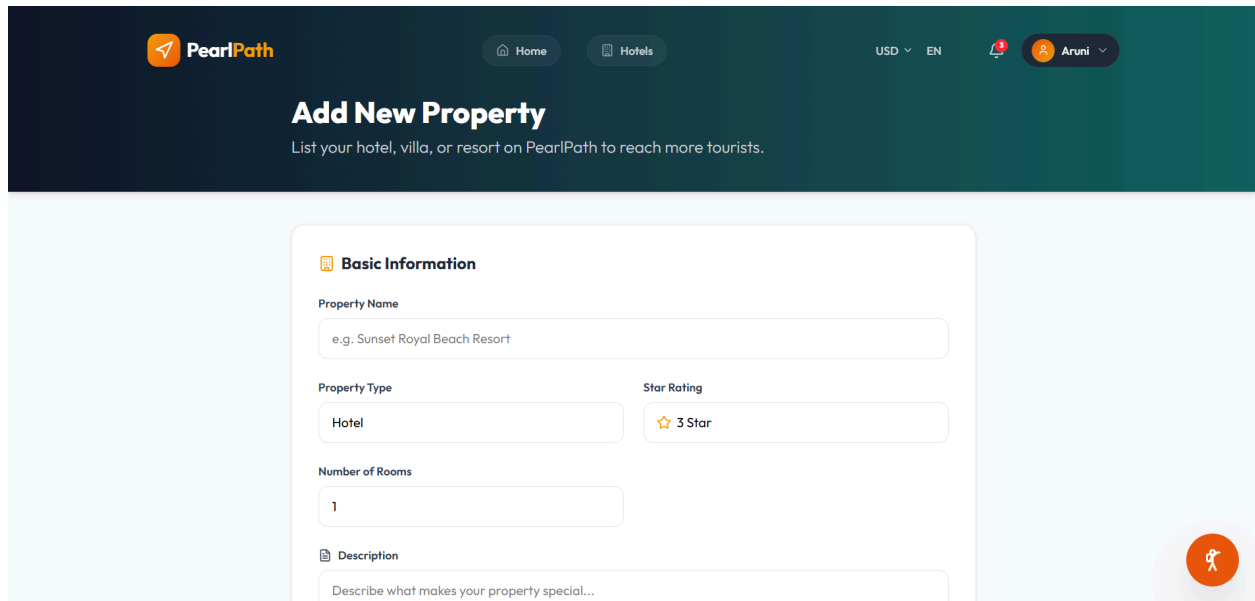
Figure 23: Provider Dashboard

6.4 Adding a Service Listing

Steps

1. Log into your Service Provider account.
2. Access the **Provider Dashboard**.
3. Click **Add New Listing**.
4. Fill in all required listing details (Title, Category, Description).
5. Upload clear, high-resolution imagery.
6. Set base pricing and precise geographical coordinates/location.

7. Outline specific amenities or operational terms.
8. Review the information and click **Save Listing**.



The screenshot shows the 'Add New Property' form on the PearlPath website. The header is dark teal with the PearlPath logo, navigation links for 'Home' and 'Hotels', currency and language selectors (USD, EN), and a user profile for 'Aruni'. The main heading is 'Add New Property' with a subtitle 'List your hotel, villa, or resort on PearlPath to reach more tourists.' The form itself is titled 'Basic Information' and contains several input fields: 'Property Name' (with a placeholder 'e.g. Sunset Royal Beach Resort'), 'Property Type' (with a dropdown menu showing 'Hotel'), 'Star Rating' (with a dropdown menu showing '3 Star'), 'Number of Rooms' (with a text input showing '1'), and 'Description' (with a placeholder 'Describe what makes your property special...'). A red circular button with a white icon is visible in the bottom right corner of the form area.

Figure 24: Add Listing Form

6.5 Updating a Service Listing

Keep service information up to date to ensure accurate customer expectations.

Steps

1. Open the **Provider Dashboard**.
2. Locate the listing card you wish to modify.
3. Click **Edit**.
4. Modify pricing, descriptions, images, or terms.
5. Click **Save Changes** and verify the update on your live preview.

6.6 Managing Availability

Service providers must update their booking calendar to reflect actual operational availability and avoid double-bookings.

Steps

1. Open the **Provider Dashboard**.
2. Select the specific service listing.
3. Navigate to the **Availability Management** tab.
4. Block out dates reserved offline or unavailable for maintenance.
5. Save changes to sync availability to the public search engine immediately.

6.7 Managing Booking Requests

Steps

1. Log into your Provider account.
2. Open **Booking Requests**.
3. Select a pending booking request to inspect reservation details.
4. Check requested date ranges and customer details.
5. Choose the appropriate response:
 - **Accept**: Confirms the booking.
 - **Decline**: Rejects the request.
6. Confirm your action. The customer receives an automatic status update.

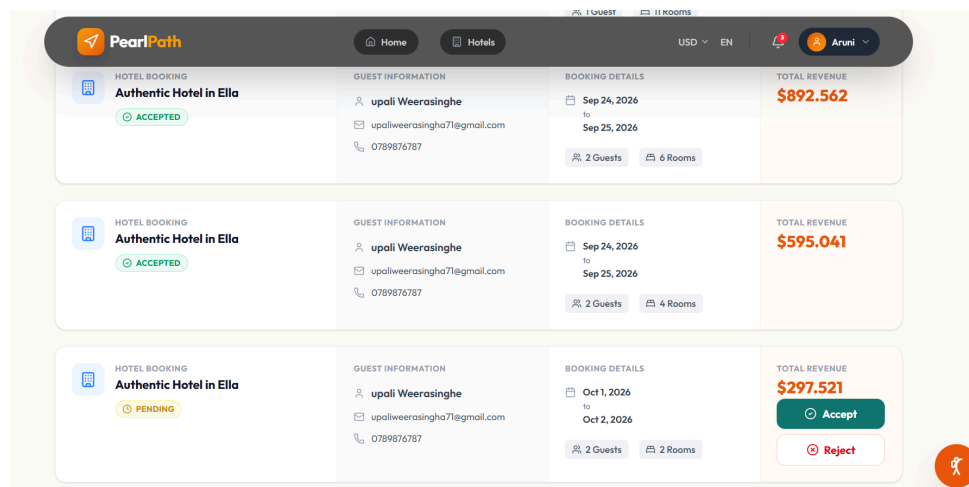


Figure 25: Provider Booking Requests

6.8 Provider Revenue and Financial Information

Service providers can review revenue summaries and financial reporting metrics directly from their control panel.

Steps

- Log into your Service Provider account.
- Open the **Provider Dashboard**.
- Click **View Financial Report** or open the revenue metrics summary block.
- Inspect confirmed total revenue, pending transaction balances, and reservation counts.

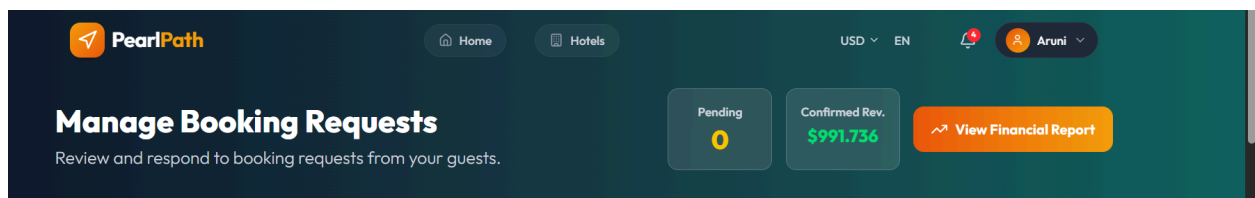


Figure 26: Provider Revenue and Financial Information

7. Administrator Guide

7.1 Administrator Dashboard

Platform administrators use the central Admin Dashboard to manage users, inspect compliance, verify providers, and monitor platform health.

Key Admin capabilities:

- User account management and role adjustments
- Service provider profile auditing and verification
- Content moderation across listings
- Platform-wide analytics and transactional oversight

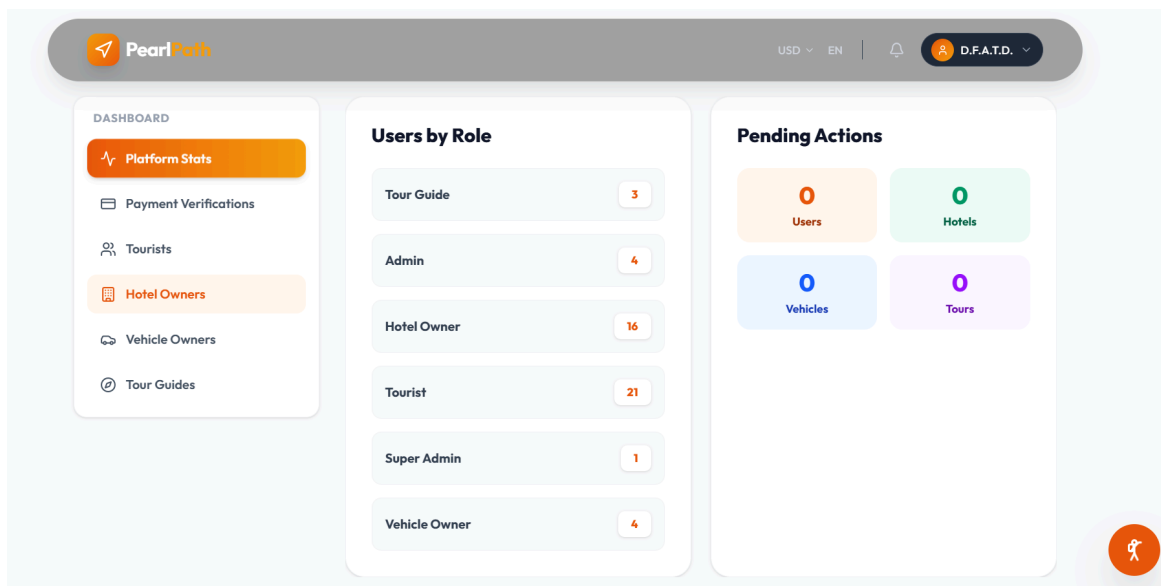


Figure 27: Administrator Dashboard

7.2 Managing Users

Steps

1. Sign in using Administrator credentials.
2. Navigate to **User Management** on the Admin Dashboard.
3. Search or filter accounts by role (Tourist, Provider, Admin).
4. Click on a user record to view account metadata.
5. Apply administrative actions as necessary (e.g., account suspension, reactivations).

7.3 Managing Service Providers

Administrators oversee service provider compliance to maintain ecosystem safety.

Steps

1. Open the Admin Dashboard.
2. Navigate to **Provider Management**.
3. Inspect registered provider accounts and associated active listings.
4. Perform necessary account edits, warnings, or removals.

7.4 Provider Verification

Steps

1. Open the Admin Dashboard.
2. Select **Pending Verifications**.
3. Open a provider application to review submitted business licenses and identity documentation.
4. Select **Approve** or **Reject** based on compliance checks.
5. Confirmed approvals unlock listing privileges for the provider.

7.5 System Monitoring

Administrators must periodically review platform health logs and transactional integrity.

Monitoring metrics include:

- Active user growth & daily active logins
- Provider listing distribution
- Pending vs. confirmed booking volumes
- System failure/error reports

8. Troubleshooting and FAQ

Q1. I cannot log into my account. What should I do?

- Check that your email address is spelled correctly.
- Verify your password (check caps lock).
- Ensure your account was verified via the email OTP step.
- Click **Forgot Password?** to reset your password.
- Check your network connection.

Q2. I did not receive my OTP verification email.

- Inspect your **Spam** or **Junk** mail folder.
- Confirm that you entered the correct email address during registration.
- Allow up to 2 minutes for email dispatch.
- Click **Resend OTP** on the verification screen.

Q3. Why is my booking attempt failing?

- The selected dates may already be booked by another user.
- Try selecting an alternative date range or browsing similar listings in that area.

Q4. Why are my desired dates blocked on the calendar?

- The provider has either blocked those dates for maintenance or another customer reservation has already been confirmed.

Q5. How can I review my active booking status?

- Navigate to **My Account** → **My Bookings** after logging in.

Q6. Why isn't a specific service appearing in search results?

- Active filter settings (e.g., price range, minimum star rating) may exclude the service. Try resetting your search filters.

Q7. The map or route calculation fails to load.

- Verify your internet connection.
- Refresh your web browser page.
- Confirm that valid, recognized location names are typed into the origin and destination inputs.

Q8. The currency conversion rate seems inaccurate.

- Currency values are updated periodically using external exchange rate integrations. They serve as informational estimates. Confirm exact rates with your bank prior to payment.

Q9. The AI Travel Assistant gives vague answers.

- Rephrase your query with specific context (e.g., specify town names, budgets, or travel style).

9. Security and Good Usage Practices

To keep your account and personal data safe, adhere to the following operational guidelines:

Account Security

- Maintain password strict confidentiality; never share account credentials.
- Use strong passwords containing uppercase letters, numbers, and symbols.
- Always log out when accessing Pearl Path from shared or public computers.

Personal Data Integrity

- Provide accurate identity details during account registration.
- Do not publish sensitive personal credentials in public review fields or query boxes.

Booking Hygiene

- Carefully verify date selections, total pricing, and location details prior to confirming bookings.

AI Assistant Guidelines

- Treat AI suggestions as general travel recommendations; independently double-check official travel notices, transit timings, and booking terms.

10. Limitations and Important Notes

Pearl Path integrates third-party services to deliver real-time functionality. Certain features may be influenced by external service operational status:

- **Maps and Route Planning:** Route calculations and map tiles rely on third-party map APIs and require an active internet connection.
- **Currency Conversion:** Foreign currency conversions are reference estimates and may not reflect real-time credit card processing fees.
- **AI Travel Assistant:** Automated suggestions generated by the travel chatbot are for guidance purposes only.
- **Service Availability:** Listing accuracy relies on service providers keeping their operational calendars up to date.
- **Email Notifications:** Delivery speed of verification OTPs and email notifications depends on external mail delivery networks.

11. Support

If you encounter technical issues that cannot be resolved using this manual, please reach out to the Pearl Path technical team.

When submitting a support ticket, please include:

1. Your account username and role (Tourist / Provider).
2. The specific page or feature where the issue occurred.
3. A step-by-step description of what caused the issue.
4. Screenshots of any error messages displayed on screen.

12. Document Version History

Version	Date	Description	Author / Team
1.0	September 2026	Initial final release of the Pearl Path User Manual & Operating Guide	Team 31 – NextDev (Department of Computer Engineering, UoP)

Pearl Path – Centralized Digital Travel Platform for Sri Lanka

Team 31 – NextDev | Department of Computer Engineering, University of Peradeniya

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